COMPUTING / COMPUTER SCIENCE

Detailed Lecture Notes

Weeks 10 – 17 | Lectures 19 – 33

Topics Covered:

Safety and Security • Modern Computing (Virtualization, Cloud, Deep/Dark Web, VR/AR, IoT)

Computer-Controlled Systems • Robotics and Automation

Digital Economics • Value Models

Entertainment Industry • Simulations

The World Wide Web • HTML, CSS, JavaScript, JSON & XML

Web Browsers and Web Servers

Each lecture includes: clear definitions, real-world examples, diagram guidance, key-point summaries, and practice questions.

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Week 10 — Lecture 19: Safety and Security

1. Introduction

Every time data is stored, transmitted, or processed on a computer system, it faces risks. 'Safety' and 'security' are related but different concepts in Computer Science, and students often confuse them in exams.

DEFINITION: *Data Security*

The protection of data from accidental or deliberate threats which could cause unauthorised amendment, disclosure, or destruction of data, and the protection of computer systems from misuse.

DEFINITION: *Data Safety*

Measures taken to protect people and equipment from harm — e.g. avoiding electric shocks, tripping hazards, or physical injury caused by hardware/wiring.

KEY POINTS TO REMEMBER

- Security = protecting data/information from unauthorised access, loss or corruption.
- Safety = protecting people and equipment from physical harm.
- Exam tip: if the question says 'harm to a person', it is a SAFETY issue; if it says 'harm to data', it is a SECURITY issue.

2. Threats to Data

Data can be threatened in many ways. These threats are usually grouped as: malicious (deliberate) threats and accidental threats.

2.1 Malware

DEFINITION: *Malware (Malicious Software)*

Software specifically written to harm, disrupt, or gain unauthorised access to a computer system.

Type	Description	Example / Effect
Virus	A program that attaches itself to a legitimate file/program and replicates when the host file is executed, spreading to other files.	Corrupts or deletes files; slows down the system.

Worm	A stand-alone program that self-replicates and spreads across networks WITHOUT needing a host file or user action.	Consumes bandwidth; can carry a 'payload' to delete files.
Trojan Horse	Malware disguised as legitimate, useful software. Does not self-replicate.	Opens a 'backdoor' allowing hackers remote access.
Spyware	Secretly monitors user activity and sends the information to a third party.	Key-logging software records every keystroke, capturing passwords.
Ransomware	Encrypts a victim's files/data and demands payment (ransom) for the decryption key.	WannaCry (2017) attack on NHS hospitals.
Adware	Automatically displays or downloads unwanted advertising material.	Pop-up adverts; can slow the system and track browsing habits.

2.2 Hacking

DEFINITION: *Hacking*

The act of gaining unauthorised access to a computer system, usually with malicious intent (to steal, change, or delete data).

EXAMPLE: Effect of Hacking

- A hacker gains access to a bank's server and transfers money out of customer accounts.
- A hacker changes exam results stored on a university's database.

2.3 Phishing

DEFINITION: *Phishing*

Sending fraudulent (fake) emails/messages that appear to come from a legitimate/trustworthy source, in order to trick the recipient into revealing personal or financial information.

EXAMPLE: Phishing scenario

- An email appears to be from your bank asking you to 'verify your account' by clicking a link and entering your card details — the link actually leads to a fake website controlled by criminals.

2.4 Pharming

DEFINITION: *Pharming*

Malicious code installed on a user's computer or on a web server that redirects the user to a fake website WITHOUT their knowledge, usually to harvest personal data.

Difference from phishing: Phishing relies on the user clicking a fake link (requires action by the victim). Pharming redirects automatically even if the correct URL is typed — it exploits DNS cache poisoning or malicious code on the device.

2.5 Social Engineering

DEFINITION: *Social Engineering*

Manipulating people psychologically into breaking normal security procedures and revealing confidential information.

- Example: A criminal phones an employee pretending to be IT support and asks for their password to 'fix an issue'.

2.6 Other Threats

Threat	Description
SQL Injection	Malicious SQL code is inserted into a data-entry field (e.g. a login box) to manipulate or access a database.

Threat	Description
Denial of Service (DoS) Attack	Flooding a server with fake requests so it cannot respond to genuine users, making a website/service unavailable.
Human Error	Accidental threats such as deleting the wrong file, spilling coffee on a device, or losing a USB stick — not deliberate, but still causes data loss.
Natural Disasters	Fire, flood, earthquake — can physically destroy hardware and stored data.

DIAGRAM: *Draw a simple flow diagram: [Attacker] → (Malware / Phishing email / Fake website) → [Victim's Computer] → (Data stolen / corrupted / encrypted). Use arrows to show the direction of the attack.*

3. Protection of Data

For every threat above, there is a corresponding protection method. It is good exam technique to pair each threat with its matching protection.

3.1 Access Control

Method	Explanation
User ID and Password	Only authorised users who know the correct login credentials can access the system.
Biometrics	Physical characteristics (fingerprint, retina scan, facial recognition) used to verify identity — hard to fake or steal.
Two-Factor Authentication (2FA)	Requires two separate methods of identification, e.g. a password AND a one-time code sent to a mobile phone.

Method	Explanation
Digital Certificates	An electronic document that proves the identity of a website/individual, issued by a trusted Certificate Authority (CA).

3.2 Encryption

DEFINITION: ***Encryption***

The process of converting data into a coded form (ciphertext) using an algorithm and a key, so that it is unreadable to anyone who does not have the correct decryption key.

DIAGRAM: *Draw: [Plaintext 'HELLO'] → (Encryption + Key) → [Ciphertext 'XQPLR'] → (Decryption + Key) → [Plaintext 'HELLO']. Label the arrow going out over a network as 'intercepted data is meaningless without the key'.*

Note: encryption does NOT stop data being intercepted — it stops the intercepted data being understood.

3.3 Firewalls

DEFINITION: ***Firewall***

Hardware or software that sits between a user's computer/internal network and an external network (e.g. the internet), examining incoming and outgoing traffic and blocking data that does not meet a set of security rules.

- Can block access to certain websites.
- Can keep a log of all incoming/outgoing traffic.
- Warns the user if suspicious activity is detected.

3.4 Anti-Malware / Anti-Virus Software

- Scans files and compares them against a database of known malware signatures.
- Quarantines or deletes infected files.
- Must be regularly updated, since new malware is created daily.

3.5 Other Protection Methods

Method	Protects Against
SSL / TLS (padlock symbol, https)	Encrypts data sent between a browser and a web server — protects online transactions.
Checking the spelling/URL of websites and sender's email address	Phishing and pharming — fake sites/emails

Method	Protects Against
	often have small errors.
Anti-spyware software	Detects and removes spyware/key-loggers.
Regular backups (on-site and off-site)	Data loss from ransomware, hardware failure, or disaster.
Staff training / awareness	Social engineering and human error.
Physical security (locks, security guards, CCTV)	Theft of hardware / unauthorised physical access.

KEY POINTS TO REMEMBER

- Every threat has a matching protection — learn them in pairs (e.g. Phishing → check sender/URL; Hacking → firewall + strong passwords).
- Encryption protects confidentiality, not availability — encrypted data can still be deleted or lost.
- A firewall controls network traffic; anti-virus software scans files already on the device — they are NOT the same thing.

PRACTICE QUESTIONS

1. Define the term 'data security' and give one example of a malicious threat to data.

2. Explain the difference between phishing and pharming.

3. Describe how encryption protects data during transmission over a network.

4. A company's firewall log shows repeated login attempts from an unknown IP address. Explain what a firewall does and why this log entry might be a cause for concern.

5. Explain the difference between a virus and a worm.

Week 10 — Lecture 20: Modern Computing

1. Virtualization and Emulation

DEFINITION: ***Virtualization***

The creation of a virtual (rather than physical) version of a computing resource, such as an operating system, server, storage device, or network, running on top of physical hardware.

DIAGRAM: *Draw a layered diagram: [Physical Hardware] → [Hypervisor] → three boxes on top labelled [Virtual Machine 1 - Windows] [Virtual Machine 2 - Linux] [Virtual Machine 3 - macOS], each running independently.*

- A Virtual Machine (VM) behaves like a separate, real computer, but actually runs inside software on a host machine.
- A hypervisor is the software that creates and manages virtual machines.

DEFINITION: ***Emulation***

Using software (an emulator) to allow one system to imitate/behave like another system, so software written for the original system can run on different hardware.

EXAMPLE: Virtualization vs Emulation

- Virtualization: Running Windows and Linux at the same time on one physical PC using VMware/VirtualBox.
- Emulation: Running old Nintendo/PlayStation games on a modern PC using a game console emulator.

Benefit	Explanation
Cost savings	One physical server can host many virtual servers, reducing hardware costs.
Easier testing	Software can be tested on a VM without risking the real/host system.
Disaster recovery	A VM can be backed up as a single file

	and quickly restored.
Better utilisation	Physical hardware resources are used more efficiently.

2. Cloud Computing

DEFINITION: ***Cloud Computing***

The delivery of computing services (storage, processing power, software, servers) over the internet ('the cloud') rather than from a local computer or on-site server.

Type of Cloud Storage	Description
Public Cloud	Storage/services owned and operated by a third-party provider and shared among multiple organisations (e.g. Google Drive, Dropbox).
Private Cloud	Storage/services used exclusively by a single organisation, either hosted internally or by a third party.
Hybrid Cloud	A combination of public and private cloud, allowing data/applications to move between the two.

Advantages and Disadvantages of Cloud Computing

Advantages	Disadvantages
Accessible from any device with an internet connection	Requires a reliable, fast internet connection
No need to buy/maintain expensive physical storage hardware	Ongoing subscription costs
Data automatically backed up by provider	Security/privacy concerns — data stored on someone else's server
Easy to scale storage up or down	Dependence on the cloud provider's reliability and policies

3. The Deep Web and Dark Web

DIAGRAM: *Draw an iceberg diagram: the small visible tip above water labelled 'Surface Web (indexed by search engines)'; the large submerged part labelled 'Deep Web (not indexed — e.g. private databases, webmail)'; a small dark section at the very bottom labelled 'Dark Web (requires special software, e.g. Tor)'.*

DEFINITION: ***Surface Web***

The part of the internet that is indexed by standard search engines (Google, Bing) and accessible to anyone using a normal browser.

DEFINITION: ***Deep Web***

The part of the internet NOT indexed by standard search engines — includes password-protected pages, private databases, academic journals, online banking pages.

DEFINITION: ***Dark Web***

A small part of the deep web that is intentionally hidden and requires special software (e.g. Tor Browser) to access; it provides anonymity for both users and website hosts.

Legitimate Uses of the Dark Web	Illegitimate Uses of the Dark Web
Journalists/whistle-blowers communicating safely under oppressive regimes	Trading illegal drugs and weapons
Privacy-conscious communication (e.g. activists, political dissidents)	Distribution of illegal/harmful material
Law enforcement/military secure communication	Hacking services and stolen data for sale

Bitcoin

DEFINITION: ***Bitcoin*** A decentralised digital cryptocurrency that can be transferred electronically between users without a central bank or single administrator, using blockchain technology to record transactions.
DEFINITION: ***Blockchain*** A distributed, public digital ledger that records transactions across many computers so that no single record can be altered without altering all subsequent records — making it very secure and transparent.

- Bitcoin is often associated with the dark web because transactions can be made with a degree of anonymity.
- Advantages: fast international transfers, low fees, no need for a bank account.
- Disadvantages: highly volatile value, used for illegal transactions, not regulated by governments, energy-intensive to 'mine'.

4. Virtual Reality (VR) and Augmented Reality (AR)

DEFINITION: ***Virtual Reality (VR)*** A completely computer-generated simulated environment that a user experiences through a headset, replacing the real world entirely with an artificial one.
DEFINITION: ***Augmented Reality (AR)*** Technology that overlays computer-generated images/information onto the real, physical world, usually viewed through a smartphone camera, tablet, or smart glasses.

Feature	Virtual Reality (VR)	Augmented Reality (AR)
Environment	Fully artificial/simulated	Real world + digital overlay
Hardware	VR headset (e.g. Oculus, HTC Vive)	Smartphone, tablet, AR glasses
Example use	Flight simulator training pilots	Pokémon GO; furniture apps showing how a sofa looks in your room

5. The Internet of Things (IoT)

DEFINITION: ***Internet of Things (IoT)***

A network of physical, everyday objects ('things') embedded with sensors, software, and network connectivity, allowing them to collect and exchange data over the internet without human intervention.

DIAGRAM: *Draw a hub-and-spoke diagram: central icon labelled [Home Wi-Fi Router / Internet] connected by lines to surrounding icons: [Smart Thermostat] [Smart Fridge] [Wearable Fitness Tracker] [Smart Doorbell/Camera] [Smart Speaker], each icon sending a small arrow labelled 'data' back to the router.*

EXAMPLE: IoT devices

- Smart thermostats that learn a household's schedule and adjust heating automatically.
- Fitness trackers that monitor heart rate and send data to a phone app.
- Smart fridges that detect low stock and can automatically add items to a shopping list.

Advantages of IoT	Disadvantages of IoT
Convenience and automation of everyday tasks	Security risk — many devices have weak security and can be hacked
Energy/cost savings (e.g. smart heating)	Privacy concerns — constant data collection
Useful data collection for health/business monitoring	Reliance on stable internet connection
Remote control/monitoring of devices	Compatibility issues between different manufacturers

KEY POINTS TO REMEMBER

- Virtualization = many virtual systems on one physical machine. Emulation = one system pretending to be a different system.
- Deep Web = simply not indexed by search engines (mostly ordinary, e.g. online banking). Dark Web = deliberately hidden, needs special software.
- VR replaces reality; AR adds to reality.
- IoT's biggest exam-relevant weakness is security, since many devices are not well protected.

PRACTICE QUESTIONS

1. Explain, using an example, what is meant by 'virtualization'.
2. Describe two differences between the deep web and the dark web.
3. Explain why Bitcoin transactions are often linked to the dark web.
4. Distinguish between virtual reality and augmented reality, giving one real-world example of each.
5. Describe two potential security risks associated with the Internet of Things.

Week 11 — Lecture 21: Computer Controlled Systems

1. What is a Computer-Controlled System?

DEFINITION: ***Computer-Controlled System***

A system where a computer, using input from sensors, makes decisions and sends signals to actuators/output devices to control a process automatically, with little or no human intervention.

DIAGRAM: *Draw a control loop diagram: [Sensor] → (sends data) → [Microprocessor/Computer] → (compares to pre-set value & makes decision) → [Actuator/Motor/Output device] → (acts on the physical process) → loop arrow back to [Sensor] labelled 'feedback loop'.*

This is called a feedback loop: the sensor constantly re-checks conditions after the actuator has acted, so the system can keep adjusting automatically.

2. Robotics in Manufacturing

DEFINITION: ***Robot***

A programmable machine capable of carrying out a series of physical actions automatically, often used to replace or assist human labour in repetitive or dangerous tasks.

EXAMPLE: Robotics in manufacturing

- Robotic arms used to weld car body panels together on an assembly line.
- Robots used to spray paint car bodies evenly, avoiding fumes that would harm human workers.
- Robots picking and packing items in a warehouse (e.g. Amazon fulfilment centres).

Feature of Robots	Explanation
Repeatability	Can perform the exact same task with the same precision, thousands of times.
Speed	Can typically work faster than a human without tiring.
Endurance	Can work continuously (24/7) without breaks, unlike human workers.

Programmability	Can be reprogrammed to perform a different task when production needs change.

3. Production Line Control

DEFINITION: ***Production Line Control***

The use of computers and sensors to automatically monitor and control each stage of a manufacturing process, ensuring consistency, quality, and efficiency.

Typical sensors used on a production line and what they monitor:

Sensor	What it Monitors	Example Use
Temperature sensor	Heat levels	Ensuring an oven baking bread stays at the correct temperature.
Pressure sensor	Force/pressure applied	Checking a bottle cap is sealed to the correct tightness.
Light/optical sensor	Presence/absence of light or objects	Counting bottles as they pass on a conveyor belt.
Weight sensor	Mass of an item	Ensuring a cereal box contains the correct weight of product.

DIAGRAM: *Draw a conveyor-belt diagram: items moving left to right past a [Weight Sensor], connected to a [Computer], which sends a signal to an [Actuator/Arm] that removes any underweight item into a 'Reject' bin.*

4. Autonomous Vehicles

DEFINITION: *Autonomous Vehicle*

A vehicle capable of sensing its environment and navigating/operating with little or no human input, using a combination of sensors, cameras, GPS, and artificial intelligence.

Component	Purpose
Cameras	Detect road markings, traffic lights, pedestrians, other vehicles.
Radar / LiDAR sensors	Measure distance to nearby objects using radio waves/laser pulses to build a 3D map of surroundings.
GPS	Determine the vehicle's exact location and plan a route.
Central Processing Computer / AI software	Processes all sensor data in real time and decides on steering, braking, and acceleration.

EXAMPLE: Autonomous vehicles

- Self-driving cars (e.g. Tesla Autopilot / Full Self-Driving features).
- Autonomous delivery drones.
- Driverless trains on some metro/underground systems.

KEY POINTS TO REMEMBER

- Computer-controlled systems rely on a constant sensor → computer → actuator → feedback loop.
- Robots in manufacturing are valued for repeatability, speed, and the ability to work in dangerous/hazardous conditions.
- Autonomous vehicles combine multiple sensor types (camera, radar/LiDAR, GPS) because no single sensor gives a complete picture.

PRACTICE QUESTIONS

1. Describe, with reference to sensors and actuators, how a computer-controlled system operates.
2. Explain two reasons why robots are used on a car manufacturing production line rather than human workers.
3. Describe the role of sensors in an autonomous vehicle, naming two types of sensor used.
4. Explain how a feedback loop allows a production line to automatically reject faulty products.

Week 11 — Lecture 22: Advantages and Disadvantages of Computer-Controlled Systems vs Humans

1. Why Compare Computers with Humans?

Exam questions frequently ask students to evaluate whether a computer-controlled system should replace human workers in a given scenario (e.g. a factory, a call centre, a hospital). A good answer considers BOTH sides and reaches a justified conclusion.

2. Advantages of Computer-Controlled Systems (over Humans)

Advantage	Explanation
Accuracy and consistency	Computers do not get tired, bored, or distracted, so they perform repetitive tasks to the same standard every time.
Speed	Computers can process information and complete physical tasks much faster than a human.
Works in hazardous environments	Can operate in extreme heat, toxic fumes, radiation, or underwater — places unsafe for humans.
No breaks needed	Can run 24 hours a day, 7 days a week, increasing productivity.
Lower long-term costs	Although expensive to install, they remove the need to pay wages, sick pay, and holiday pay over time.
No human error from fatigue	Reduces mistakes caused by tiredness, distraction, or emotional stress.

3. Disadvantages of Computer-Controlled Systems (compared to Humans)

Disadvantage	Explanation
High initial/setup cost	Buying and installing robots/automated systems is very expensive.
Lack of flexibility/judgement	Computers can only do what they are programmed to do; they struggle with unexpected situations requiring human judgement or creativity.
Unemployment	Automation can replace human jobs, leading to job losses and the need for staff retraining.
Maintenance and technical faults	Requires skilled technicians to repair; a technical fault can halt an entire production line.
Lack of empathy/emotional intelligence	Unsuitable for jobs requiring emotional understanding, e.g. caregiving, counselling, customer complaint handling.
Security vulnerabilities	Computer-controlled systems connected to a network can be hacked, unlike a purely human-run process.

4. Balanced Evaluation — Example Scenario

EXAMPLE: Sample Evaluation: Should a hospital replace nurses with robots for patient care?

- FOR: Robots could accurately deliver medication doses and monitor vital signs continuously without fatigue.
- AGAINST: Patient care requires empathy, reassurance, and the ability to respond to unpredictable emotional/medical situations — something robots cannot yet replicate.
- CONCLUSION: A hybrid approach is often best — robots assist with repetitive/monitoring tasks (e.g. dispensing medicine, moving supplies) while humans remain responsible for direct patient care and decision-making.

KEY POINTS TO REMEMBER

- Always give BOTH advantages and disadvantages, then a justified conclusion tailored to the scenario given in the question.
- Common exam trap: don't just say 'computers are faster' — explain WHY that matters in the given context (e.g. faster production = more products made per day = increased profit).
- Jobs requiring empathy, creativity, or complex judgement are the strongest argument AGAINST full automation.

PRACTICE QUESTIONS

1. Discuss the advantages and disadvantages of replacing human workers with robots on a car production line.

2. A supermarket is considering replacing its human checkout staff with fully automated self-checkout systems. Discuss the advantages and disadvantages of this decision.

3. Explain why full automation may not be suitable for jobs such as counselling or childcare.

4. Evaluate whether autonomous vehicles are safer than human-driven vehicles. Justify your answer.

Week 12 — Lecture 23: Digital Economics

1. Digital Goods and Services

DEFINITION: *Digital Goods* Intangible assets/products that exist only in digital form and are distributed electronically, such as e-books, music files, software, and streaming content.
DEFINITION: *Digital Services* Services delivered over the internet or a digital network rather than in person, such as online banking, video streaming, or cloud storage.

Feature	Digital Goods	Physical Goods
Reproduction cost	Very low — near zero cost to duplicate	High — requires materials, manufacturing
Distribution	Instant, via download/streaming	Requires shipping/transport, takes time
Storage	No physical space needed	Requires warehouse/shelf space
Example	An e-book, an MP3 file, a software licence	A printed book, a CD, a boxed software copy

2. Production of Digital Services

There are several different models by which digital goods/services are produced. Each has its own structure and motivation.

2.1 In-House Production

DEFINITION: *In-House Production* When an organisation uses its own employees and resources to develop a digital product/service internally, rather than outsourcing it to another company.

Advantages	Disadvantages
Full control over quality and direction	Higher cost — must employ specialist staff
Better protection of trade secrets/intellectual property	Slower — limited to the

	skills/capacity of existing staff
Easier to tailor the product to exact company needs	Risk if key staff leave the company

2.2 Commons-Based Peer Production

DEFINITION: ***Commons-Based Peer Production***

A model of production where large numbers of people work together voluntarily over the internet, without traditional hierarchical management or direct financial reward, to create a shared resource.

EXAMPLE: Commons-based peer production

- Wikipedia — thousands of volunteers write and edit articles collaboratively, free for anyone to use.
- Open-source software projects like Linux, where programmers worldwide contribute code voluntarily.

- Motivation for contributors: reputation, personal satisfaction, learning, ideology (belief in free/shared knowledge) rather than direct payment.

2.3 Crowdsourcing

DEFINITION: ***Crowdsourcing***

Obtaining input, ideas, content, services, or funding by requesting contributions from a large group of people, typically via the internet, rather than from traditional employees or suppliers.

Type of Crowdsourcing	Example
Crowdfunding	Kickstarter/GoFundMe campaigns where the public funds a new product or project.
Crowd-sourced data/content	Waze traffic app, where drivers report traffic conditions in real time.
Crowd-sourced labour	Amazon Mechanical Turk, where many people complete small online tasks.

DIAGRAM: *Draw a hub diagram: a central box [Company/Project] surrounded by many small stick-figure icons labelled 'The Crowd', each with an arrow pointing INTO the central box labelled 'ideas / funding / data / tasks'.*

2.4 Open-Source Software

DEFINITION: ***Open-Source Software***

Software whose source code is made publicly available, allowing anyone to view, use, modify, and distribute it, usually under a licence that requires modifications to remain open too.

Open-Source Software	Proprietary (Closed-Source) Software
Source code is publicly available	Source code is kept secret/hidden
Usually free to use and modify	Usually requires payment/licence fee
Developed collaboratively by a community	Developed and controlled by a single company
Example: Linux, LibreOffice, Mozilla Firefox	Example: Microsoft Windows, Microsoft Office, Adobe Photoshop

Advantages of Open-Source Software	Disadvantages of Open-Source Software
Usually free to use	May lack dedicated/guaranteed technical support
Can be customised/modified to suit specific needs	Documentation can be inconsistent or incomplete
Transparent — code can be checked for security flaws by anyone	Compatibility issues with some proprietary software/hardware
Large community support/forums	Quality can vary between different open-source projects

KEY POINTS TO REMEMBER

- In-house production = controlled internally by paid staff. Commons-based peer production = voluntary, collaborative, no central company control.
- Crowdsourcing specifically means gathering contributions (ideas/funding/data/labour) from a large external 'crowd'.
- Open-source refers to the code being publicly viewable/modifiable — it does NOT always mean free of charge, though it very often is.

PRACTICE QUESTIONS

1. Define the term 'digital goods' and give two examples.
2. Explain, using an example, what is meant by 'commons-based peer production'.
3. Describe two differences between open-source and proprietary software.
4. Explain what is meant by 'crowdsourcing', giving one real-world example.
5. Discuss one advantage and one disadvantage of a company using in-house production to develop new software.

Week 12 — Lecture 24: Value Models

1. Introduction to Value Models

A 'value model' describes how an organisation creates, delivers, and captures value for its customers. Different types of business use different value models depending on how they operate.

2. Value Chain

DEFINITION: ***Value Chain***

A model describing a business as a linear sequence of activities, each of which adds value to a product as it moves from raw material to finished product delivered to the customer.

DIAGRAM: *Draw a horizontal chain of connected boxes (left to right): [Inbound Logistics] → [Operations] → [Outbound Logistics] → [Marketing & Sales] → [Service]. Underneath, draw a second row of 'Support Activities' boxes spanning the whole chain: [Firm Infrastructure] [HR Management] [Technology Development] [Procurement].*

Best suited to businesses that transform inputs into outputs in a fixed, sequential order — such as manufacturing.

EXAMPLE: Value chain example

- A car manufacturer: raw steel/parts arrive (inbound logistics) → parts assembled into a car (operations) → finished cars shipped to dealerships (outbound logistics) → advertising to attract buyers (marketing/sales) → warranty repairs (service).

3. Value Shop

DEFINITION: ***Value Shop***

A model describing a business that creates value by solving unique customer problems, mobilising resources and expertise for each specific case rather than following a fixed, repeated sequence.

DIAGRAM: *Draw a circular/cyclic diagram (not linear): [Problem Identification] → [Diagnosis] → [Solution Design] → [Execution] → [Evaluation/Control] → arrow looping back to [Problem Identification], since each case may require revisiting earlier stages.*

EXAMPLE: Value shop example

- A hospital: each patient presents a unique problem; doctors diagnose, design a treatment plan, carry it out, then evaluate results — the process can loop back if the first treatment doesn't work.
- A law firm handling an individual client's legal case.
- A software consultancy solving a specific client's IT problem.

4. Value Network

DEFINITION: ***Value Network***

A model describing a business that creates value by connecting and enabling exchanges/relationships between customers or clients, rather than transforming inputs (value chain) or solving problems (value shop).

DIAGRAM: *Draw a network/web diagram: a central box labelled [Platform / Network Provider] with multiple two-way arrows connecting to surrounding user icons, showing users connected to EACH OTHER through the central platform, not just to the provider.*

EXAMPLE: Value network example

- A telecommunications company (mobile network provider) — its value comes from connecting millions of subscribers to each other.
- Social media platforms like Facebook — value comes from connecting users, not from manufacturing a product.
- A bank — connects savers with borrowers.

5. Comparing the Three Value Models

Model	Core Logic	Structure	Example Industry
Value Chain	Transform inputs into outputs	Linear, sequential	Manufacturing (e.g. cars, electronics)
Value Shop	Solve unique problems	Cyclical, iterative	Professional services (hospitals, law firms, consultancies)
Value Network	Connect people/customers to each other	Networked, mediating	Telecoms, banks, social media platforms

KEY POINTS TO REMEMBER

- Value Chain = STEP-BY-STEP transformation of raw materials into a product (think: factory line).
- Value Shop = PROBLEM-SOLVING for a unique case (think: doctor, lawyer, consultant).
- Value Network = CONNECTING two or more parties together (think: telecom, bank, social media).
- Exam tip: to identify which model applies, ask 'does this business make a physical product (chain), solve individual problems (shop), or

Model	Core Logic	Structure	Example Industry
connect people/parties together (network)?'			
PRACTICE QUESTIONS 1. Define the term 'value chain' and give an example of a business that fits this model. 2. Explain why a hospital is better described using the value shop model rather than the value chain model. 3. Describe, using an example, what is meant by a 'value network'. 4. A mobile telecommunications company wants to describe how it creates value for its customers. Identify and justify which value model best fits this business.			

Week 13 — Lecture 25: Entertainment Industry

1. The Animation Industry

DEFINITION: ***Animation***

The process of creating the illusion of movement by displaying a rapid sequence of still images (frames) that differ slightly from one another.

Type of Animation	Description	Example
Traditional (2D hand-drawn) Animation	Each frame is drawn individually by hand (or digitally in 2D) and played in sequence.	Classic Disney films (e.g. early Disney cartoons).
3D / CGI Animation	Computer software is used to build 3D models, which are then 'rigged' (given a digital skeleton) and animated by manipulating that skeleton.	Pixar films such as Toy Story.
Stop-Motion Animation	Physical objects/models are photographed one frame at a time, being moved slightly between each shot.	Wallace and Gromit.
Motion Capture (MoCap)	Sensors are placed on a real actor's body to record their movement digitally, which is then applied to an animated 3D character.	Gollum in The Lord of the Rings; many modern video game characters.

DIAGRAM: *Draw a simple 'flip-book' diagram: 4-5 small stick-figure frames side by side, each slightly different in pose, with an arrow underneath labelled 'played quickly in sequence = illusion of movement (frame rate, e.g. 24 frames per second)'.*

Impact of Computers on the Animation Industry

Benefit	Explanation
Efficiency	Computers can automatically generate 'in-between' frames (tweening) instead of an artist drawing every single frame by hand.
Realism	3D rendering and lighting engines can produce highly realistic visuals not possible by hand.
Reusability	Digital character models/rigs can be reused and adjusted for sequels or different scenes.
Special effects	Complex effects (explosions, water, crowds) can be simulated using physics engines.

2. The Game Industry

DEFINITION: ***Video Game***

An electronic game that involves interaction with a user interface or input device to generate visual feedback on a display, often built using a 'game engine'.

DEFINITION: ***Game Engine***

A software framework that provides the core tools needed to build a video game, such as a rendering engine (graphics), physics engine, sound engine, and scripting tools — e.g. Unity or Unreal Engine.

Roles in Game Development

Role	Responsibility
Game Designer	Plans gameplay mechanics, rules, storylines, and levels.
Programmer	Writes the code that makes the game function (physics, AI, controls).
Artist / Animator	Creates visual assets: characters, environments, and animations.
Sound Designer/Composer	Creates sound effects and music for the game.
Tester (QA)	Plays the game to find and report bugs before release.

Impact of the Games Industry

Positive Impacts	Negative Impacts / Concerns
Major economic contributor — a multi-billion-dollar global industry	Concerns about gaming addiction, especially among young people
Drives advances in graphics/hardware technology (GPUs)	Health issues from prolonged sitting/screen time
Provides entertainment, social connection (online multiplayer), and even education (serious games)	Concerns over violent content and its effects, especially on children
Esports has created new career and entertainment opportunities	In-game purchases/'loot boxes' criticised as encouraging gambling-like behaviour

KEY POINTS TO REMEMBER

- Know the four main animation types: traditional (2D), 3D/CGI, stop-motion, motion capture — and be able to name an example of each.
- A game engine (e.g. Unity, Unreal) provides ready-made tools (graphics, physics, sound) so developers don't have to build everything from scratch.
- For 'impact' questions, always give both positive AND negative points.

PRACTICE QUESTIONS

1. Describe two different types of animation technique, giving an example of each.

2. Explain what is meant by a 'game engine' and name one example.

3. Discuss the positive and negative impacts of the video game industry on society.

4. Explain how motion capture technology is used to create realistic animated characters.

Week 13 — Lecture 26: Simulations and Their Uses

1. What is a Simulation?

DEFINITION: *Simulation*

A computer program that creates a model of a real-world system or process, allowing it to be tested and studied under different conditions without the cost, risk, or time of using the real thing.

Simulations are built from mathematical models representing real-world variables and their relationships. Because they use models rather than the real system, they only produce results as accurate as the model itself.

2. Why Use Simulations?

Reason	Explanation
Safety	Allows testing of dangerous scenarios (e.g. plane crashes, nuclear reactor faults) without real risk to life.
Cost	Cheaper than building/testing multiple real physical prototypes.
Time	Can compress processes that take years (e.g. climate change) into minutes of computer processing.
Repeatability	The same scenario can be run repeatedly, changing one variable at a time, to study cause and effect.
Impossible/impractical scenarios	Allows study of situations that cannot be recreated physically, such as future population growth.

3. Examples of Simulations

Simulation Type	Real-World Use
Flight simulators	Training pilots to handle emergencies (e.g. engine failure) safely, without risking a real aircraft.
Weather/climate simulations	Predicting weather patterns and modelling long-term

Simulation Type	Real-World Use
	climate change.
Medical simulations	Training surgeons on virtual patients before operating on real people; modelling the spread of diseases.
Crash test simulations	Testing car safety in a collision without destroying a real vehicle each time.
Economic/financial simulations	Modelling the effect of interest rate changes on a national economy.
Urban planning simulations	Modelling traffic flow to design better road networks and reduce congestion.

DIAGRAM: *Draw an input-process-output diagram: [Real-world variables entered as input data, e.g. speed, weight, weather conditions] → [Computer model processes the data using mathematical rules] → [Output: predicted results, e.g. graphs, visual animation, statistics].*

4. Advantages and Disadvantages of Simulations

Advantages	Disadvantages
Safe way to study dangerous situations	A simulation is only as accurate as the

Advantages	Disadvantages
	model/data used to build it — an inaccurate model gives inaccurate/misleading results
Saves money compared to real-world testing	Can be very expensive and time-consuming to develop initially
Allows repeated testing by changing variables	May not account for every real-world variable, so results can differ from reality
Speeds up processes that would take a long time in reality	Users may over-rely on simulation results without questioning their accuracy

KEY POINTS TO REMEMBER

- A simulation is only ever a MODEL of reality — its accuracy depends entirely on how good and complete that model is.
- The main reasons to simulate: safety, cost, time, repeatability, and studying otherwise impossible scenarios — learn these five.
- Always be ready to link a simulation example to a specific real-world reason it is useful (e.g. flight simulator → safety).

PRACTICE QUESTIONS

1. Define the term 'simulation'.

2. Explain two reasons why an airline would use a flight simulator to train pilots rather than a real aircraft.

3. Discuss one advantage and one disadvantage of using computer simulations to predict climate change.

4. Explain why the accuracy of a simulation's results depends on the model used to create it.

Week 14 — Lecture 27: The World Wide Web — Overview, Distributed Web, Addressable Web

1. The Internet vs The World Wide Web

Students very commonly confuse these two terms — this distinction is a favourite exam question.

DEFINITION: ***The Internet***

A global network of interconnected computer networks that use standardised communication protocols (TCP/IP) to exchange data. It is the physical/logical infrastructure.

DEFINITION: ***The World Wide Web (WWW)***

A system of interlinked hypertext documents (web pages) accessed via the internet, using web browsers. The Web is a SERVICE that runs on top of the internet.

The Internet	The World Wide Web
The physical network of connected computers/cables/routers	A collection of web pages/documents accessible via the internet
Existed before the Web (since ~1960s-70s)	Invented later by Tim Berners-Lee in 1989-1991
Includes email, file transfer (FTP), VoIP calls, the Web, and more	Is just ONE of the many services that runs on the internet

DIAGRAM: *Draw a large circle labelled [The Internet]. Inside it, draw several smaller separate circles labelled [World Wide Web], [Email], [Instant Messaging], [File Transfer/FTP], [Online Gaming], showing that the Web is just one service running on the internet, not the same thing as the internet.*

2. The Distributed Web

DEFINITION: ***Distributed Web***

The concept that the World Wide Web's content is not stored in one single central location, but is instead spread across millions of independent servers located all over the world.

- No single organisation owns or controls the whole Web.
- Each website's files are hosted on its own web server (or servers), which may be located anywhere in the world.
- This distributed nature makes the Web resilient — if one server goes down, only that website is affected, not the entire Web.

DIAGRAM: *Draw a world map outline with 5-6 small server icons scattered across different continents, each connected by lines through a cloud labelled [The Internet], showing that web content is physically stored in many different, independent locations.*

3. The Addressable Web

DEFINITION: ***Addressable Web***

The principle that every resource (web page, image, document) on the Web has a unique address, allowing it to be located and retrieved directly.

DEFINITION: ***URL (Uniform Resource Locator)***

A unique text address used to locate a specific resource on the Web, specifying the protocol, domain name, and path to the resource.

DIAGRAM: *Draw a labelled breakdown of a URL, e.g.: https://www.example.com/products/laptop.html — label each part underneath: 'https' = Protocol, 'www.example.com' = Domain name (identifies the server), '/products/laptop.html' = Path (location of the specific file on that server).*

Part of URL	Example	Purpose
Protocol	https://	States the rules used to transfer the data (secure HTTP).
Domain Name	www.example.com	Identifies which web server to connect to (translated to an IP address via DNS).
Path	/products/laptop.html	Identifies the specific file/resource on that server.

Because every resource has its own unique URL, users (and search engines) can link directly to a specific page rather than always starting from a website's homepage — this is what makes the Web 'addressable'.

KEY POINTS TO REMEMBER

- The Internet is the infrastructure; the Web is a service (of hyperlinked documents) that runs on that infrastructure.
- 'Distributed' = content is stored across many independent servers worldwide, not in one place.
- 'Addressable' = every single resource has its own unique URL, so it can be found and linked to directly.

PRACTICE QUESTIONS

1. Explain the difference between the Internet and the World Wide Web.
2. Explain what is meant by describing the Web as 'distributed'.
3. Identify and describe the three main parts of the URL: https://www.school.edu/notes/week10.pdf
4. Explain why the addressable nature of the Web is important for search engines.

Week 14 — Lecture 28: The Linked Web, Protocols, Searchable Web, Languages of the Web

1. The Linked Web

DEFINITION: ***Linked Web (Hypertext)***

The concept that web pages are connected to one another through hyperlinks, allowing users to navigate from one document to another regardless of where each document is physically stored.

DEFINITION: ***Hyperlink***

A clickable reference (usually text or an image) within a web page that, when selected, takes the user to another web page, resource, or location within the same page.

DIAGRAM: *Draw a node-and-line 'web' diagram: 6-7 circles labelled Page A, Page B, Page C, etc., connected to each other with arrows representing hyperlinks — some pages linking to several others, showing a non-linear, interconnected structure (hence 'World Wide WEB').*

This linked structure is precisely why the system is called a 'Web' — pages connect to each other in a complex network rather than a simple linear order.

2. The Protocols of the Web

DEFINITION: ***Protocol***

An agreed set of rules that governs how data is formatted, transmitted, and received between devices on a network.

Protocol	Full Name	Purpose
HTTP	HyperText Transfer Protocol	The standard protocol used to transfer web pages/data between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	The same as HTTP but data is encrypted using SSL/TLS, protecting it from interception.
FTP	File Transfer Protocol	Used to upload/download files between a client and a server.

TCP/IP	Transmission Control Protocol / Internet Protocol	The core set of protocols that manages how data is broken into packets, addressed, routed, and reassembled across the internet.
SMTP / IMAP / POP3	Simple Mail Transfer Protocol / Internet Message Access Protocol / Post Office Protocol	Used for sending (SMTP) and receiving (IMAP/POP3) email.

EXAMPLE: HTTP request-response cycle

- 1. Browser sends an HTTP request to a web server asking for a page (e.g. GET /index.html).
- 2. Web server processes the request and locates the file.
- 3. Server sends an HTTP response back containing the requested page's HTML code (plus a status code, e.g. 200 OK, or 404 Not Found).
- 4. Browser renders (displays) the HTML as a visible web page.

3. The Searchable Web

DEFINITION: *Search Engine*

A software system designed to search the World Wide Web for pages matching a user's query, using an index built by automated programs.

Stage	Description
1. Crawling	Automated programs called 'web crawlers' (or 'spiders'/'bots') systematically browse the Web, following hyperlinks from page to page.

2. Indexing	The content found by crawlers is analysed and stored in a huge searchable database called an index.
3. Ranking / Retrieval	When a user types a search query, the search engine's algorithm ranks matching pages from its index by relevance, and displays results.
DIAGRAM: *Draw a 3-step flow diagram: [Web Crawler visits pages via hyperlinks] → arrow → [Pages added to Search Index (database)] → arrow → [User types query] → arrow → [Ranking Algorithm] → arrow → [List of results shown to user].*	

- Ranking factors can include: keyword relevance, number/quality of links pointing to a page, page loading speed, and how frequently the content is updated.

4. The Languages of the Web

Web pages are built using several different types of language, each with a distinct purpose. This links directly into the following lectures (HTML, CSS, JavaScript, JSON/XML) which will be covered in full detail.

Language	Type	Purpose	Characteristics
HTML (HyperText Markup Language)	Markup language	Defines the STRUCTURE and content of a web page (headings, paragraphs, images, links).	Not a programming language — no logic/calculations; uses 'tags' to mark up content.
CSS (Cascading Style Sheets)	Style sheet language	Controls the PRESENTATION/appearance of a web page (colours, fonts, layout, spacing).	Separates content (HTML) from design, allowing one style sheet to control many pages.

Language	Type	Purpose	Characteristics
JavaScript	Scripting/programming language	Adds INTERACTIVITY and dynamic behaviour to a web page (e.g. responding to clicks, validating forms).	Runs mainly in the browser (client-side); supports variables, loops, functions, and logic.
JSON (JavaScript Object Notation)	Data-interchange format	Used to STORE and TRANSPORT structured data between a server and a web page.	Lightweight, human-readable, uses key–value pairs; easy for both humans and machines to parse.
XML (eXtensible Markup Language)	Markup / data-interchange format	Used to STORE and TRANSPORT structured data, similar in purpose to JSON but with custom tags.	More verbose than JSON; strictly hierarchical, with custom-defined tags similar to HTML.

EXAMPLE: Analogy to remember the languages

- HTML = the skeleton/bricks of a house (structure).
- CSS = the paint, wallpaper, and furniture (appearance).
- JavaScript = the electricity and switches (behaviour/interactivity).
- JSON/XML = the moving vans that transport furniture (data) between different houses (systems).

KEY POINTS TO REMEMBER

- Hyperlinks create the 'web' shape of the World Wide Web — a non-linear network of connected documents.

Language	Type	Purpose	Characteristics
• HTTP transfers web pages; HTTPS is the encrypted, secure version; TCP/IP is the underlying protocol suite for the whole internet. • Search engines work in three stages: crawling → indexing → ranking/retrieval. • Remember the 5 languages of the web and their ONE main purpose each: HTML=structure, CSS=style, JavaScript=behaviour, JSON/XML=data transport.			
PRACTICE QUESTIONS 1. Explain what is meant by a 'hyperlink' and why it makes the Web a 'linked' system. 2. Describe the difference between HTTP and HTTPS. 3. Describe the three stages involved in a search engine finding and displaying results for a user's query. 4. Identify the language of the web that would be used to: (a) structure a page's headings and paragraphs, (b) change the background colour of a page, (c) validate that a user has entered a valid email address in a form. 5. Explain one similarity and one difference between JSON and XML.			

Week 15 — Lecture 29: Structuring the Web with HTML

1. What is HTML?

DEFINITION: *HTML (HyperText Markup Language)*
The standard markup language used to create and structure the content of web pages, using a system of 'elements' marked by tags.

HTML is NOT a programming language — it has no logic, variables, or calculations. It simply describes the structure and meaning of content (e.g. 'this text is a heading', 'this is a paragraph', 'this is a link').

2. HTML Tags and Elements

DEFINITION: *Tag*

A keyword surrounded by angle brackets (< >) that marks the beginning or end of an HTML element, e.g. <p> and </p>.

DEFINITION: *Element*

A complete unit consisting of an opening tag, its content, and a closing tag, e.g. <p>This is a paragraph.</p>.

DEFINITION: *Attribute*

Extra information added inside an opening tag to modify an element's behaviour or provide additional data, written as name="value".

EXAMPLE: Anatomy of an element

- Click here
- <a> is the opening tag, is the closing tag, 'Click here' is the content, and href="..." is an attribute specifying the link's destination.

3. Basic Structure of an HTML Document

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Web Page</title>
  </head>
  <body>
    <h1>Welcome to My Page</h1>
    <p>This is a paragraph of text.</p>
  </body>
</html>
```

Part	Purpose
<!DOCTYPE html>	Tells the browser this document

	uses HTML5.
<html>...</html>	The root element — contains the entire web page.
<head>...</head>	Contains information ABOUT the page (title, links to CSS files) — not shown directly on the page.
<title>...</title>	Sets the text shown in the browser tab/title bar.
<body>...</body>	Contains all the VISIBLE content of the web page.
DIAGRAM: *Draw a nested-box diagram: outer box labelled <html>, containing two boxes side by side labelled <head> (with a small note 'title, meta info') and <body> (with a small note 'headings, paragraphs, images, links — visible content').*	

4. Common HTML Tags

Tag	Purpose	Example
<h1> to <h6>	Headings (h1 = largest/most important, h6 = smallest)	Main Title
<p>	Paragraph of text	Some text.
<a>	Hyperlink (anchor)	Next Page
<img>	Embeds an image (self-closing tag)	<img src="photo.jpg" alt="A photo">
/ /	Unordered (bulleted) / ordered (numbered) list, containing list items	Item 1

Tag	Purpose	Example
<div>	A generic block-level container, used to group content for styling/layout	<div class="box">...</div>
	A generic inline container, used to style a small part of text	Warning
<table>, <tr>, <td>	Creates a table, table row, and table cell	<table><tr><td>Data</td></tr></table>
<form>, <input>, <button>	Creates an interactive form to collect user input	<input type="text">

5. Attributes in Detail

Attribute	Used On	Purpose
href	<a>	Specifies the destination URL of a hyperlink.
src	<img>	Specifies the source file (path/URL) of an image.
alt	<img>	Provides alternative text if the image cannot load, and helps accessibility/screen readers.
id	any element	Gives an element a unique identifier, used by CSS/JavaScript to target it specifically.
class	any element	Assigns an element to a named group, allowing CSS to style multiple elements together.

6. A Simple Complete Example

```
<!DOCTYPE html>
<html>
<head>
  <title>About Me</title>
</head>
<body>
  <h1>About Me</h1>
  <img src="profile.jpg" alt="My profile picture">
  <p>Hello! I am a student learning web development.</p>
  <h2>My Hobbies</h2>
  <ul>
    <li>Reading</li>
    <li>Coding</li>
    <li>Gaming</li>
  </ul>
  <a href="contact.html">Contact Me</a>
</body>
</html>
```

KEY POINTS TO REMEMBER

- HTML defines STRUCTURE and content only — it has no logic and is not a programming language.
- Most elements have an opening AND closing tag; a few (like <img>) are 'self-closing' / 'void' elements.
- 'id' should be unique per page (one element only); 'class' can be reused on many elements.
- The <head> section is never shown directly to the user; the <body> section holds everything visible.

PRACTICE QUESTIONS

1. Explain why HTML is described as a 'markup language' rather than a 'programming language'.
2. Write the HTML code needed to display a level-1 heading that says 'My Website'.
3. Explain the difference between the 'id' and 'class' attributes.
4. Write HTML code to create a hyperlink that says 'Visit Google' and links to https://www.google.com.
5. Describe the purpose of the <head> section of an HTML document, giving one example of content it might contain.

Week 15 — Lecture 30: Styling the Web with CSS

1. What is CSS?

DEFINITION: ***CSS (Cascading Style Sheets)***

A style sheet language used to control the presentation and visual appearance of a web page written in HTML, including colours, fonts, spacing, and layout.

CSS separates a page's CONTENT (HTML) from its DESIGN (CSS). This means one CSS file can control the appearance of an entire website, and the design can be changed without touching the HTML content at all.

2. Ways to Add CSS to a Web Page

Method	Description	Example
Inline CSS	Written directly inside an HTML element's 'style' attribute. Affects only that one element.	Text
Internal (Embedded) CSS	Written inside a <style> tag within the <head> of the HTML document. Affects that one page only.	<style>p { color: blue; }</style>
External CSS	Written in a separate .css file and linked to the HTML document. Can control MANY pages from one file — the recommended method.	<link rel="stylesheet" href="style.css">

DIAGRAM: *Draw a diagram showing one file [style.css] with arrows pointing to three separate boxes: [index.html] [about.html] [contact.html] — illustrating how a single external CSS file can style multiple HTML pages at once.*

3. CSS Syntax

```
selector {
  property: value;
  property: value;
}

/* Example */
h1 {
  color: navy;
  font-size: 32px;
  text-align: center;
}
```

Term	Meaning
Selector	Identifies WHICH HTML element(s) the rule applies to (e.g. h1, p, .classname, #idname).
Property	The aspect of the element being styled (e.g. color, font-size, margin).
Value	The setting applied to that property (e.g. navy, 32px, center).
Declaration	One property : value; pair.
Declaration Block	The full set of declarations inside the curly braces { }.

4. Types of CSS Selectors

Selector Type	Syntax	Targets
Element selector	p { }	Every <p> element on the page.
Class selector	.highlight { }	Every element with class="highlight".
ID selector	#header { }	The single element with id="header".
Descendant selector	div p { }	Every <p> that is inside a <div>.

5. The 'Cascading' in Cascading Style Sheets

DEFINITION: *Cascade*

The set of rules that determine which CSS style applies when multiple conflicting rules could apply to the same element — based on specificity, source order, and importance.

General rule of priority (highest to lowest): inline CSS > ID selector > class selector > element selector. If two rules have equal priority, the one written LAST in the code wins.

EXAMPLE: Cascade example

- If a CSS file sets 'p { color: black; }' but also sets '.warning { color: red; }', then <p class="warning">Text</p> will appear RED, because the class selector is more specific than the element selector.

6. The Box Model

DEFINITION: *Box Model*

The concept that every HTML element is treated as a rectangular box, made up of four layers: content, padding, border, and margin.

DIAGRAM: *Draw the CSS box model as nested rectangles from inside out, each labelled: [Content] innermost, surrounded by [Padding], surrounded by [Border], surrounded by [Margin] (outermost, transparent). Label margin as 'space outside the border, between this box and other boxes'.*

Layer	Description
Content	The actual text/image inside the element.
Padding	Space between the content and the border (inside the box).

Border	A line surrounding the padding and content.
Margin	Space outside the border, separating this element from neighbouring elements.

7. Example: HTML + CSS Together

```
<!-- HTML -->
<link rel="stylesheet" href="style.css">
<h1 class="title">Welcome</h1>
<p id="intro">This is my website.</p>

/* CSS (style.css) */
.title {
  color: darkblue;
  font-family: Arial, sans-serif;
}

#intro {
  font-size: 18px;
  padding: 10px;
  border: 1px solid grey;
}
```

KEY POINTS TO REMEMBER

- CSS controls presentation only; it never changes what the content IS, only how it looks.
- External CSS is best practice because one file can style an entire website.
- Selector priority (specificity): inline > ID > class > element. Last rule wins if specificity is equal.
- The box model order (inside → out) is: Content → Padding → Border → Margin. A common mnemonic is 'Cats Peacefully Bring Milk'.

PRACTICE QUESTIONS

1. Explain the difference between internal and external CSS.
2. Write a CSS rule that makes all <h2> headings red and centred.
3. Explain what is meant by the 'cascade' in CSS, using an example.
4. Describe, using a diagram, the four layers of the CSS box model.
5. A web page has a rule 'p { color: black; }' and a rule '#note { color: green; }'. Explain which colour a paragraph with id="note" would display, and why.

Week 16 — Lecture 31: Scripting the Web with JavaScript

1. What is JavaScript?

DEFINITION: *JavaScript*

A scripting/programming language used to make web pages interactive and dynamic, by allowing code to run in response to user actions and to change page content without reloading the page.

HTML	CSS	JavaScript
Structure/content	Presentation/style	Behaviour/interactivity
The skeleton	The appearance	The actions/logic

Unlike HTML and CSS, JavaScript IS a full programming language: it supports variables, decisions (if statements), loops, and functions — meaning it can perform calculations and make decisions.

2. Where JavaScript Runs

DEFINITION: *Client-Side Scripting*

Code that runs in the user's own web browser (on their device) rather than on the web server. JavaScript is traditionally a client-side language.

- Because it runs on the user's device, client-side JavaScript can respond instantly (e.g. validating a form) without waiting for a round-trip to the server.
- Node.js is a technology that allows JavaScript to also run on the SERVER side, but this is beyond this course's core focus.

3. Adding JavaScript to a Web Page

```
<!-- Internal JavaScript -->
<script>
  alert('Hello, World!');
</script>

<!-- External JavaScript -->
<script src="script.js"></script>
```

4. Core JavaScript Concepts

Concept	Description	Example
Variable	A named storage location for a value that can change.	let score = 0;
Data types	The kind of value stored: number, string (text), boolean (true/false), array, object.	"Hello" is a string; 42 is a number; true is a boolean

Concept	Description	Example
Operator	A symbol used to perform an operation on values.	+, -, *, /, ==, &&
Condition (if statement)	Runs code only if a specified condition is true.	if (score > 10) { ... }
Loop	Repeats a block of code multiple times.	for (let i = 0; i < 5; i++) { ... }
Function	A reusable, named block of code that performs a specific task.	function greet() { alert('Hi'); }
Event	An action detected by the browser (e.g. a click, a key press) that can trigger a function.	button.onclick = function() {...}

5. The Document Object Model (DOM)

DEFINITION: ***DOM (Document Object Model)***

A programming interface/tree-like representation of an HTML document, which allows JavaScript to access, read, and change the content, structure, and style of a web page dynamically.

DIAGRAM: *Draw a tree diagram: top node [document] branching down to [html], which branches to [head] and [body]; [body] branches further down to [h1], [p], [ul], and [ul] branches to two [li] nodes. Label this 'The DOM Tree'.*

JavaScript uses the DOM to 'find' an element (e.g. by its id) and then change it — for example, changing text, hiding an element, or reacting to a button click.

6. Practical Example

```
<!-- HTML -->
<button id="myButton">Click Me</button>
<p id="message"></p>

<!-- JavaScript -->
<script>
  function showMessage() {
    document.getElementById('message').innerHTML = 'Button was clicked!';
  }
  document.getElementById('myButton').onclick = showMessage;
</script>
```

EXAMPLE: What happens step by step

- 1. The browser loads the HTML page, including the button and empty paragraph.
- 2. JavaScript finds the button element using its id ('myButton') via the DOM.
- 3. An event listener is attached: when the button is clicked, the showMessage() function runs.
- 4. The function finds the paragraph with id 'message' and changes its content — the page updates instantly, with no reload.

7. Common Uses of JavaScript on Websites

Use	Example
Form validation	Checking a password field is not empty before the form is submitted.
Dynamic content updates	Displaying a live countdown timer without reloading the page.
Interactive elements	Image sliders/carousels, drop-down menus, pop-up notifications.
Fetching data	Retrieving data from a server (often in JSON format) and displaying it without a full page reload (AJAX).

KEY POINTS TO REMEMBER

- JavaScript is a real programming language (unlike HTML/CSS) — it can store data, make decisions, and repeat actions.
- Client-side means it runs in the user's browser, giving instant responses without contacting the server each time.
- The DOM is the 'bridge' that allows JavaScript to read and change HTML content and structure while the page is running.
- Events (like onclick) are what connect a user's action to a JavaScript function.

PRACTICE QUESTIONS

1. Explain the difference between HTML, CSS, and JavaScript in terms of what each one controls.
2. Explain what is meant by 'client-side scripting'.
3. Describe what the DOM is and why it is important for JavaScript.
4. Write simple JavaScript code (or pseudocode) that displays an alert box saying 'Welcome' when a button is clicked.
5. Explain why form validation is often carried out using JavaScript rather than relying only on the server.

Week 16 — Lecture 32: Structuring the Web's Data with JSON and XML

1. Why Do We Need Data-Interchange Formats?

Websites and apps constantly need to send structured data between a server and a browser (or between two different systems) — for example, a weather app fetching today's forecast, or an online shop fetching a product's price and stock level. JSON and XML are the two most common formats used for this purpose.

2. JSON (JavaScript Object Notation)

DEFINITION: ***JSON***

A lightweight, text-based, human-readable data-interchange format that represents data as key–value pairs, structured similarly to a JavaScript object.

```
{
  "name": "Ahmed",
  "age": 21,
  "isStudent": true,
  "subjects": ["Computing", "Mathematics", "Physics"]
}
```

JSON Element	Meaning
{ }	Curly braces represent an OBJECT — a set of key–value pairs.
[ ]	Square brackets represent an ARRAY — an ordered list of values.
"key": value	A property name (in quotes) paired with its value, separated

	by a colon.
,	Separates multiple key–value pairs or array items.

3. XML (eXtensible Markup Language)

DEFINITION: ***XML***

A markup language designed to store and transport data, using custom-defined tags to describe the structure and meaning of the data, similar in appearance to HTML.

```
<student>
  <name>Ahmed</name>
  <age>21</age>
  <isStudent>true</isStudent>
  <subjects>
    <subject>Computing</subject>
    <subject>Mathematics</subject>
    <subject>Physics</subject>
  </subjects>
</student>
```

Unlike HTML, XML tags are NOT predefined — the author creates their own tag names (e.g. <student>, <name>) to describe their specific data.

4. Comparing JSON and XML

Feature	JSON	XML
Syntax style	Key–value pairs using { } and [ ]	Nested custom tags, similar to HTML
Readability	Generally shorter and easier to read	More verbose (needs opening AND closing tags for everything)
File size	Typically smaller	Typically larger due to repeated tags
Data types supported	Supports numbers,	Everything is text by default;

Feature	JSON	XML
	strings, booleans, arrays, objects natively	data type must be interpreted separately
Common use	Most modern web APIs and JavaScript applications	Configuration files, older/enterprise systems, documents (e.g. RSS feeds)
Comments allowed?	No	Yes (<!-- comment -->)

DIAGRAM: *Draw a side-by-side comparison box: LEFT box titled 'JSON' showing a small key-value tree; RIGHT box titled 'XML' showing the same data as a nested-tag tree — both representing identical data ('name: Ahmed') to visually show they can express the same information differently.*

5. Why Structured Data Formats Matter

Reason	Explanation
Interoperability	Allows systems built in different programming languages to exchange data using a shared, agreed format.
Human readability	Both formats can be read and understood directly by a person, unlike binary data formats.
Easy to parse	Both have well-defined rules, so software

Reason	Explanation
	libraries can reliably convert the text into usable data structures.
API communication	Most web APIs (e.g. a weather service) return data in JSON format for an app to process and display.

KEY POINTS TO REMEMBER

- JSON uses key–value pairs and is based on JavaScript object syntax — generally more compact than XML.
- XML uses custom, self-describing tags in a nested structure similar to HTML — allows comments, which JSON does not.
- Both formats exist to let different systems exchange structured data in a way that is both human-readable and machine-parsable.
- Modern web APIs mostly favour JSON, but XML remains common in older/enterprise systems and certain document formats (e.g. RSS).

PRACTICE QUESTIONS

1. Explain the purpose of a data-interchange format such as JSON or XML.
2. Represent the following data as JSON: a book with title 'Computing Basics', 300 pages, and available = true.
3. Represent the same book data (title, pages, available) as XML.
4. Describe two differences between JSON and XML.
5. Explain why JSON is commonly preferred over XML in many modern web applications.

Week 17 — Lecture 33: Web Browsers and Web Servers

1. Web Browsers

DEFINITION: *Web Browser*

Software application that allows a user to access, request, and display content from the World Wide Web, by sending requests to web servers and rendering the responses received.

EXAMPLE: Common web browsers

- Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge, Opera

1.1 Functions of a Web Browser

Function	Explanation
Requesting pages	Sends an HTTP/HTTPS request to a web server, asking for a specific resource (identified by a URL).
Rendering	Interprets HTML, CSS, and JavaScript code received from the server and displays it visually as a formatted web page.
Running scripts	Executes client-side JavaScript to add interactivity to the page.
Managing history/bookmarks	Keeps track of previously visited pages and allows users to save favourite pages.
Managing cookies	Stores small pieces of data sent by websites, used to remember login state, preferences, or shopping cart contents.
Security checks	Verifies SSL/TLS certificates for HTTPS sites and warns users about unsafe/insecure websites.

1.2 The Rendering Engine

DEFINITION: *Rendering Engine*

The part of a web browser responsible for translating HTML and CSS code into the visual layout that the user sees on screen.

- Different browsers use different rendering engines (e.g. Blink in Chrome, Gecko in Firefox, WebKit in Safari) — this is why the same website can occasionally look slightly different across browsers.

2. Web Servers

DEFINITION: *Web Server*

A computer (hardware) or software application that stores website files and responds to requests from web browsers by sending back the requested web pages/resources.

EXAMPLE: Common web server software

- Apache HTTP Server, Nginx, Microsoft Internet Information Services (IIS)

2.1 Functions of a Web Server

Function	Explanation
Storing website files	Holds all the HTML, CSS, JavaScript, image, and other files that make up a website.
Listening for requests	Waits for incoming HTTP/HTTPS requests from browsers (clients) on a specific port (usually 80 for HTTP, 443 for HTTPS).
Processing requests	Locates the requested file (or generates dynamic content, e.g. by querying a database) based on the URL received.
Sending responses	Returns the requested resource to the browser, along with an HTTP status code indicating success or failure.
Handling multiple clients	Can serve requests from many different users/browsers simultaneously.

3. The Client-Server Model: How a Browser and Server Communicate

DEFINITION: ***Client-Server Model***

A distributed system architecture in which a client (e.g. a web browser) requests services/resources, and a server responds by providing them.

DIAGRAM: *Draw a step-by-step client-server diagram: [Client / Web Browser] --1. HTTP Request (URL typed/link clicked)--> [Web Server]; [Web Server] --2. Locates requested file / runs server-side script--> [Web Server]; [Web Server] --3. HTTP Response (HTML, CSS, JS, images + status code)--> [Client / Web Browser]; [Client / Web Browser] --4. Renders the page on screen.*

3.1 Step-by-Step: Loading a Web Page

Step	What Happens
1	User types a URL into the browser (or clicks a hyperlink).
2	The browser uses DNS (Domain Name System) to translate the domain name into the web server's IP address.
3	The browser sends an HTTP/HTTPS request to that IP address, asking for the specific resource in the URL's path.
4	The web server receives the request, locates (or generates) the requested file.
5	The web server sends an HTTP response back, containing the requested HTML/CSS/JS/images and a status code.
6	The browser's rendering engine processes the HTML/CSS and runs any JavaScript, displaying the finished page to the user.

3.2 Common HTTP Status Codes

Code	Meaning	Example
200	OK — the request was successful.	The page loaded correctly.

Code	Meaning	Example
301 / 302	Redirect — the resource has moved to a different URL.	www.site.com redirecting to www.site.com/home
404	Not Found — the requested resource does not exist on the server.	A broken link/mistyped URL.
500	Internal Server Error — something went wrong on the server itself.	A fault in server-side code/database.

4. Static vs Dynamic Web Pages

Static Web Page	Dynamic Web Page
Content is fixed — the same for every visitor, stored as a ready-made HTML file	Content can change based on user, time, or data — generated by the server on request
Faster to serve (no processing needed)	Requires server-side processing (e.g. database queries) before sending a response
Example: a simple informational 'About Us' page	Example: a social media feed, or an online shop showing live stock/prices

KEY POINTS TO REMEMBER

- Browser = client software that REQUESTS and DISPLAYS (renders) web pages. Server = stores files and RESPONDS to requests.
- The client-server model is the foundation of how the entire Web operates: request → process → response → render.

Static Web Page	Dynamic Web Page
● DNS translates a human-readable domain name into the IP address needed to actually locate the server. ● Know at least the meanings of HTTP status codes 200, 404, and 500 — these come up frequently in exams.	
PRACTICE QUESTIONS 1. Explain the roles of a web browser and a web server in displaying a web page to a user. 2. Describe, in order, the steps that occur between a user typing a URL and the web page appearing on their screen. 3. Explain the purpose of DNS in this process. 4. State what is meant by the HTTP status codes 404 and 500, giving an example of when each might occur. 5. Explain the difference between a static and a dynamic web page, giving one example of each.	